CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
First Semester Master of Technology (CE)
University Examination, January 2011
CE707: Distributed Computing

Date: 07/01/2011 **Time: 10:30 a.m. to 1:30 p.m.** **Maximum Marks: 70**

Instructions:
1. Figure to the right indicates full marks
2. Make suitable assumption whenever necessary and mention it clearly
3. Parts of same question should be answered together and in the same sequence.
4. Answers to the sections must be written separately.

SECTION-I

Q-1 **Answer the followings:**

[A] What type of infrastructure is required by Cloud computing? Explain limitations of Cloud computing in brief. [03]

[B] What is Cluster computing? Explain limitations of Cluster computing. [02]

[C] What is XML Schema? Explain with example. [02]

[D] What is significance of WSDL? Explain use of disco file. [04]

Q-2 **Attempt any three:** [12]

1. Write a code to create and consume XML web service in .Net for User login process. (use VB or C# as language)
2. Define term Web session. Explain use of Hidden Form Fields and Cookies for transferring session state data.
3. Explain HTTP protocol events and HTTP message harder with diagram.
4. Explain layout of a SOAP response with source code format. Briefly describe each element of it.

Q-3 **Attempt the following questions:**

[A] What is Cloud computing? Explain Cloud Computing architecture and any one industry frameworks. [04]

[B] Define terms in context of Grid computing: Meta-scheduling, Monitoring. [02]

[C] Explain concept of ActiveX EXE, write a code on which takes input as group of numbers and return sorted number group using ActiveX EXE. Write steps to use this ActiveX EXE in other application. [06]

OR

[A] What is Grid computing? Explain Gird Architecture with diagram. [04]

[B] Discuss practical applications of cloud computing for various industries with example. [02]

[C] Explain concept of ActiveX DLL, write a code for ActiveX DLL which takes input as any one arithmetic operator and two operands and return result of this operation. Write steps and code for an application which use this ActiveX DLL, [06]

SECTION-II

Q-4 Attempt following:

1. Explain four basic primitive operations needed for IPC. [04]
2. Describe java RMI architecture. What is role of RMI registry? [04]
3. Explain java IDL. [03]

Q-5

[A] Write a java code fragment that may appear in main method to open datagram socket to receiving datagram of up to 100 bytes, timeout in 5 seconds. If timeout does occur, a message "Time-out on receive" should be display on screen. [06]

[B] Explain key methods and constructor of Socket class. What is use of SocketImpl in stream programming? [06]

OR

[B] Write a java code for day-time multicast client/server application.

Q-6 Write short notes: (Any Three) [12]

1. Text-based protocol
2. ORB
3. Secure socket layer
4. RMI security manager